

Ring switching, generalized

How to transform a PCS for $\mathbb{F}_{2^f}$ into a PCS for $\mathbb{F}_{2^e}$, for arbitrary e and f .

Goal. We want f *bit*-valued multilinear $P_1, \dots, P_f : \{0, 1\}^m \rightarrow \{0, 1\}$ over the same m variables, to be committed through a dense PCS over the *packing* field $\mathbb{F}_{2^f}$ (if we can commit to bits, we can trivially commit in any extension). The evaluation claims, however, arrive over a *different* field: at an opening point $r \in \mathbb{F}_{2^e}^m$ we are given $P_i(r) = \alpha_i \in \mathbb{F}_{2^e}$. Ring switching [1] packs the f polynomials into one $\mathbb{F}_{2^f}$ -dense polynomial and reduces the f claims to a single claim on it. The two extension degrees, f (packing) and e (opening), are unrelated (not necessarily a sub-field / extension-field).

Packing. Fix an $\mathbb{F}_2$ -basis $\{b_i^F\}_{i=1}^f$ of $\mathbb{F}_{2^f}$. Consider the multilinear polynomial:

$$P_{\text{packed}}(y) = \sum_{i=1}^f P_i(y) b_i^F, \quad y \in \{0, 1\}^m,$$

dense over $\mathbb{F}_{2^f}$, which we commit to.

Reduction. The claims share a *common* point $r \in \mathbb{F}_{2^e}^m$; each is

$$\alpha_i = P_i(r) = \sum_{y \in \{0, 1\}^m} \text{eq}(r, y) P_i(y),$$

with bits $P_i(y)$ and weights $\text{eq}(r, y) \in \mathbb{F}_{2^e}$. Fix an $\mathbb{F}_2$ -basis $\{b_w^E\}_{w=1}^e$ of the *opening* field $\mathbb{F}_{2^e}$ and expand the weights, $\text{eq}(r, y) = \sum_{w=1}^e A(y, w) b_w^E$, with $A : \{0, 1\}^m \times [e] \rightarrow \mathbb{F}_2$ public, multilinear in y for each fixed w , with efficiently computable MLE (see [Evaluating \$B\$](#) below). Substituting and pulling out the basis,

$$\alpha_i = \sum_{w=1}^e b_w^E \underbrace{\sum_{y \in \{0, 1\}^m} A(y, w) P_i(y)}_{\in \mathbb{F}_2},$$

so the w -th $\mathbb{F}_2$ -coordinate of α_i is $\alpha_{w,i} = \sum_{y \in \{0, 1\}^m} A(y, w) P_i(y)$. Now recombine *across* i with the *packing* basis $\{b_i^F\}$: define, for each $w \in \{1, \dots, e\}$,

$$t_w := \sum_{i=1}^f \alpha_{w,i} b_i^F = \sum_{y \in \{0, 1\}^m} A(y, w) \underbrace{\sum_{i=1}^f P_i(y) b_i^F}_{= P_{\text{packed}}(y)} = \sum_{y \in \{0, 1\}^m} A(y, w) P_{\text{packed}}(y) \in \mathbb{F}_{2^f}$$

one claim per w , now purely about the packed polynomial P_{packed} over $\mathbb{F}_{2^f}$ and public A . Batch these e claims with powers of a random $\gamma \leftarrow \mathbb{F}_{2^f}$ into a single sumcheck on $\sum_{y \in \{0, 1\}^m} B(y) P_{\text{packed}}(y) = \sum_{w=1}^e \gamma^w t_w$, with $B(y) = \sum_{w=1}^e \gamma^w A(y, w)$. Its output is the packed claim $P_{\text{packed}}(r') = \alpha'$ at a point $r' \in \mathbb{F}_{2^f}^m$ (plus a $B(r')$ the verifier computes itself). Done.

Soundness. Suppose the input is wrong: $\alpha_i \neq P_i(r)$ for some i , against the committed P_{packed} . Then the reduction outputs a correct packed claim $P_{\text{packed}}(r') = \alpha'$ with probability at most

$$\underbrace{\frac{e}{2^f}}_{\text{batching}} + \underbrace{\frac{2m}{2^f}}_{\text{sumcheck}} = \frac{2m + e}{2^f},$$

Proof:

1. The map $(\alpha_i)_{i=1}^f \mapsto (t_w)_{w=1}^e$ is a *bijection* $\mathbb{F}_{2^e}^f \rightarrow \mathbb{F}_{2^f}^e$. So a wrong input forces, with certainty, $t_w \neq \sum_{y \in \{0,1\}^m} A(y, w) P_{\text{packed}}(y)$ for some w .
2. By Schwartz–Zippel, $\sum_{w=1}^e \gamma^w t_w$ then differs from $\sum_{y \in \{0,1\}^m} B(y) P_{\text{packed}}(y)$ except with probability $\leq e/2^f$ over $\gamma \leftarrow \mathbb{F}_{2^f}$.
3. Sumcheck started from this false value (summand of degree 2, over m variables) ends at a correct claim $P_{\text{packed}}(r') = \alpha'$ with probability $\leq 2m/2^f$.

Evaluating B

It remains to compute $B(r')$, where $B(y) = \sum_{w=1}^e \gamma^w A(y, w)$ and $A(y, w) \in \mathbb{F}_2$ is the w -th coordinate of $\text{eq}(r, y)$, i.e. $\text{eq}(r, y) = \sum_{w=1}^e A(y, w) b_w^E$. Following [2], this gives B a closed form on the hypercube. Let $\Phi : \mathbb{F}_{2^e} \rightarrow \mathbb{F}_{2^f}$ be the $\mathbb{F}_2$ -linear map with $\Phi(b_w^E) = \gamma^w$. For every $y \in \{0, 1\}^m$:

$$\Phi(\text{eq}(r, y)) = \Phi\left(\sum_{w=1}^e A(y, w) b_w^E\right) = \sum_{w=1}^e A(y, w) \Phi(b_w^E) = \sum_{w=1}^e A(y, w) \gamma^w = B(y).$$

So on the hypercube B is “compute $\text{eq}(r, y) \in \mathbb{F}_{2^e}$, then apply Φ ”.

Now $\text{eq}(r, y) = \prod_{j=1}^m g_j(y_j)$ with $g_j(0) = 1 - r_j$, $g_j(1) = r_j$, and multiplication by a fixed element of $\mathbb{F}_{2^e}$ is an $e \times e$ matrix $M_j^{(y_j)}$ over $\mathbb{F}_2$; so, on $\{0, 1\}^m$, B is a width- e *matrix* branching program: the product $\prod_j M_j^{(y_j)}$ (started from the coordinates of 1), followed by Φ . Its multilinear extension is then evaluable at r' in $O(me^2)$ field operations (see [3, Section 4.1], or more generally [4]).

Remarks

1. *Small $\mathbb{F}_{2^f}$.* If 2^f is too small for the error $(2m + e)/2^f$, sample γ (and run the sumcheck) in an extension $L \supseteq \mathbb{F}_{2^f}$: the error becomes $(2m + e)/|L|$, and P_{packed} is just opened at $r' \in L^m$.
2. *Any field.* Nothing uses characteristic 2: replace $\mathbb{F}_2$ by any field K and $\mathbb{F}_{2^f}, \mathbb{F}_{2^e}$ by degree- f and degree- e extensions of K ; it still works.

Credits

The idea comes from Lev Soukhanov, `[[alloc]init]`.

References

- [1] Benjamin E. Diamond and Jim Posen. Polylogarithmic proofs for multilinears over binary towers. In *Advances in Cryptology – EUROCRYPT 2026*, pages 3–32. Springer, 2026.
- [2] Benedikt Bünz, Ron Rothblum, and William Wang. Flock: Fast proving for batch Boolean computations. Cryptology ePrint Archive, Paper 2026/1329, 2026. <https://eprint.iacr.org/2026/1329>.
- [3] Tamir Hemo, Kevin Jue, Eugene Rabinovich, Gyumin Roh, and Ron D. Rothblum. Jagged polynomial commitments (or: How to stack multilinears). Cryptology ePrint Archive, Paper 2025/917, 2025. <https://eprint.iacr.org/2025/917>.
- [4] Justin Holmgren and Ron D. Rothblum. Delegating computations with (almost) minimal time and space overhead. In *59th IEEE Annual Symposium on Foundations of Computer Science, FOCS 2018*, pages 124–135. IEEE, 2018.